Carbon Sequestration Ecosystem Service Accounts

European Union, 2000 to 2024 series update: processing methods and results

Renato Vargas
Consultant for JRC

Alessandra La Notte
Senior Environmental Economist at European Dynamics

Table of contents

Summary 2

1 Introduction 2

2 Methods 4
2.1 Overview . . . 4
2.2 Removals and emissions . . . 4
2.2.1 Harvested wood products . . . 5
2.3 The offset and the actual flow . . . 5
2.3.1 Choice of the allocation key . . . 6
2.4 Allocation to emitting sectors . . . 7
2.5 Regional values . . . 7
2.6 Monetary valuation . . . 7
2.7 Maps . . . 8
2.7.1 The spatial key . . . 8
2.7.2 Products . . . 9
2.8 Checks . . . 9

3 Results 11
3.1 Removals, emissions and actual flow . . . 11
3.2 Countries . . . 12
3.3 Regions . . . 14
3.4 Maps . . . 15
3.5 Value . . . 15
3.6 Sectors . . . 19
3.7 Comparison with the published accounts . . . 20

4 Limitations 21

References 21

Annex I. Scripts of the pipeline 23

Summary

We provide estimates of the carbon sequestration ecosystem service for the 27 Member States of the European Union in 2000, 2006, 2012, 2018 and 2024, following the accounting method of Vallecillo et al. (2019) and its update in La Notte et al. (2021). In 2024, land removes 375.0 million tonnes of CO2 and emits 139.1 million tonnes. The actual flow that reaches society after this offset is 236.0 million tonnes, against 326.3 million tonnes in 2000, and it is worth 16.3 billion euro at constant 2010 prices. The missed flow, which is the part of the removals used up by those emissions, is worth 9.6 billion euro. The value rests on a social cost of carbon of EUR 343.6 per tonne of carbon, which is EUR 93.7 per tonne of CO2, in 2023 euros.

1 Introduction

Carbon sequestration is the uptake of carbon dioxide (CO2) from the atmosphere by ecosystems, mainly through the growth of trees, which stores carbon in biomass and soil. Ecosystem accounting treats it as a contribution to global climate regulation that benefits society as a whole. Ecosystems also release CO2, so the accounts need to separate what land removes from what land emits, and to establish how much of the removals is left once those emissions are compensated (La Notte et al., 2021).

This report describes how we estimated those quantities and valued them, why we took each decision along the way and what the estimates show. The calculations are carried out by a series of scripts in the folder https://github.com/renatovargas/jrc-carbon-sequestration/tree/main/scripts, listed in Table 1 in the order in which they run. Each step of the Methods section points to the entry of its script in Annex I, which explains what the script does and links to its code. The file run_all.R runs the whole series or any of its stages, and any single script can run again when only one input changes. The tables and the 100 m maps can be retrieved from the Downloads page of this site.

Table 1: Scripts of the pipeline, in the order in which they run.

Script	Purpose
01_prepare_nuts.R	Country boundaries
02_download_eurostat.R	CO2 inventory by source sector
03_download_hicp.R	Consumer price index

Script	Purpose
04_prepare_clc.R	Forest share at 1 km and land type areas by region
05_prepare_gdmp.R	Annual productivity at 1 km
06_prepare_nuts_weights.R	Share of each region in the weight of each land type
07_prepare_carbon_price.R	Carbon price series
08_build_physical_accounts.R	Removals, emissions, actual flow, missed flow and sector allocation
09_build_monetary_accounts.R	Value of the flows, regional results and the workbook
10_write_maps_1km.R	Maps at 1 km
11_write_maps_100m.R	Maps at 100 m
12_check_maps_100m.R	Comparison of the 100 m maps with the tables

The terms in Table 2 recur throughout the report.

Table 2: Key terms of the report.

Term	Meaning in this report
LULUCF	Land use, land use change and forestry, the part of the greenhouse gas inventory that reports the CO2 exchanged between land and the atmosphere.
Removal	CO2 taken up by a land category in a country and year, when the category is a net sink.
Emission	CO2 released by a land category in a country and year, when the category is a net source.
Offset	The netting of all emissions of land against all removals, at the level of the European Union.
Actual flow	The net CO2 that ecosystems deliver to society after the offset.
Flow without offset	The total removals of the land categories, before their emissions are subtracted.
Missed flow	The flow without offset minus the actual flow, which is the part of the removals that compensates the emissions of land use.
GDMP	Gross Dry Matter Productivity, the rate at which vegetation produces dry matter, estimated from satellite images.
NUTS	The European classification of territorial units, where NUTS0 is the country level.

2 Methods

2.1 Overview

National greenhouse gas inventories report CO2 by country, and they do not report it by place. The accounts therefore work in two stages. They first build the national and regional values, which is a statistical problem, and they then place the national values on a map, which is a spatial problem. Monetary values follow from the physical ones through a price per tonne.

The analysis covers the EU27, with 239 NUTS2 regions and 1160 NUTS3 regions under the NUTS 2024 boundaries (Eurostat GISCO, 2024), and maps use the projection EPSG:3035, in which every cell covers the same area on the ground (see 01_prepare_nuts.R in Annex I). The boundaries carry the notice © EuroGeographics for the administrative boundaries.

2.2 Removals and emissions

We took CO2 by source sector from the Eurostat dataset `env_air_gge` (Eurostat, 2026b), in the release retrieved on 2026-10-30, for the 27 Member States and the EU27 aggregate (see 02_download_eurostat.R in Annex I). Following Vallecillo et al. (2019), the land categories are forest land, cropland, grassland, wetlands, settlements and other land, which are the codes `CRF4A` to `CRF4F`.

The inventory uses the point of view of the atmosphere, so a negative value is CO2 removed and a positive value is CO2 released. The accounts reverse the sign, so that a positive balance b of category k in country c is uptake. The removals R and emissions E are the two parts of that balance.

$$R_{k,c} = \max(b_{k,c}, 0), \qquad E_{k,c} = \max(-b_{k,c}, 0) \tag{1}$$

A category that is a net source in a year therefore counts fully as an emission and adds nothing to the removals of that year. Forest land is a net source in several countries, as Table 3 shows for the reporting years. A category that a country does not report is set to zero and listed. Among the six land categories, this concerns only other land in Denmark, in 25 years. The residual category `CRF4H`, which 14 of the 27 countries report, stays outside the account. The calculations of this section and the next two are carried out by the same script (see 08_build_physical_accounts.R in Annex I).

Table 3: Countries where forest land is a net source of CO2 in the reporting years, in thousand tonnes of CO2.

Year	Country	Balance of forest land (kt CO2)
2018	DE	-6,911
2018	EE	-1,778
2018	AT	-1,028
2018	SI	-968
2018	CZ	-843
2024	AT	-2,642
2024	DE	-1,648
2024	PT	-261

2.2.1 Harvested wood products

The inventory reports the change in the stock of harvested wood products as its own category, `CRF4G`. A deduction of that sink from the removals of forest land is one possible treatment, since the carbon in harvested wood has left the forest. The carbon chapters of Vallecillo et al. (2019) and La Notte et al. (2021) take forest uptake from the inventory as reported and make no such deduction, and the handbook of Eurostat keeps these products out of net sequestration as a separate category (Eurostat, 2026a, para. 9.56). A decision was therefore made not to deduct them. The sink stays outside the flow as a memo item, at 24.8 million tonnes in 2024.

2.3 The offset and the actual flow

La Notte et al. (2021) define the service that reaches society as the balance of removals and emissions over all ecosystem types at the level of the European Union, so an ecosystem type with more emissions than removals provides no service. The EU net flow is computed once.

$$N = \sum_{k,c} R_{k,c} - \sum_{k,c} E_{k,c} \tag{2}$$

La Notte et al. (2021) then attribute this net flow to each country c in proportion to its removals of forest land $R_{F,c}$, in their Equation 6.1.

$$F_c = N \frac{R_{F,c}}{\sum_c R_{F,c}} \tag{3}$$

The flow without the offset W_c of a country is the sum of its removals over the six land categories, and the missed flow M_c is the difference between that and the actual flow.

$$W_c = \sum_k R_{k,c}, \qquad M_c = W_c - F_c \tag{4}$$

At the level of the European Union, the missed flow equals total emissions by construction. The definition follows from the distinction that La Notte et al. (2021) draw between the role of forests, shown by removals, and the contribution to society, shown by the actual flow.

2.3.1 Choice of the allocation key

Net annual increment, which is the growth of living trees minus natural mortality, is another candidate for the key in Equation 3, because it measures forest growth and does not depend on the sign of the inventory. We compared the two approaches in a run that used net annual increment of all forest from Eurostat (Eurostat, 2026d) and also deducted harvested wood products (Table 4). That run assigned a flow to countries whose forest land was a net source, including 26.1 million tonnes to Germany in 2024, and it left between 7 and 12 countries with a negative missed flow in each reporting year, which has no direct reading as a service that is not delivered. The series of net annual increment also carried breaks between years, and very few countries had reported it for 2024, so most of the key of that year would have repeated earlier values.

A decision was made to follow Equation 6.1. Under this key no country has a negative missed flow, and a country whose forest land is a net source receives no flow, which agrees with the definition of the service. The comparison was run once, so the pipeline does not repeat it, and the table keeps its result. The sink in that table adds up the countries where the category is a sink, so it is slightly larger than the net balance of the category that the text above and Table 6 report.

Table 4: EU27 actual flow under the method of this report and under the compared run, in million tonnes of CO2, with the number of countries with a negative missed flow.

Year	Actual flow, this report	Actual flow, compared run	Harvested wood products sink	Negative missed flow, this report	Negative missed flow, compared run
2000	326.3	280.9	46.3	0	7

Year	Actual flow, this report	Actual flow, compared run	Harvested wood products sink	Negative missed flow, this report	Negative missed flow, compared run
2006	355.1	292.6	62.9	0	12
2012	371.7	340.0	31.7	0	8
2018	225.2	194.2	40.2	0	9
2024	236.0	213.0	25.8	0	8

2.4 Allocation to emitting sectors

Vallecillo et al. (2019) complement the accounts with a table that attributes the flow of each country to the economic sectors in proportion to their share of its total CO2 emissions, which shows which sectors the service offsets. The allocation does not change the flow. We grouped the source categories of the inventory into energy supply, manufacturing and construction, transport, agriculture, forestry and fishing, households, services and other, and waste. These keep the sector names of the inventory, which differ from the industries of the national accounts, as the fuel that households burn in their cars is reported under transport.

2.5 Regional values

NUTS2 and NUTS3 regions receive a share of the national values (see 06_prepare_nuts_weights.R in Annex I for the shares, and 09_build_monetary_accounts.R in Annex I for the regional results). The actual flow of a region is the national flow times the share of the region in the forest weight of its country, which is the spatial key described below. The flow without the offset of a region adds up the removals of the six land categories, each shared in proportion to the area of that category in the region according to the Corine accounting layer of the year. The missed flow of a region is the difference between the two. When a country has no mapped area of a category, the share of the region in the total mapped land of the country stands in. This concerns only LU and MT.

2.6 Monetary valuation

The value of a flow is the flow in tonnes of CO2 multiplied by a price per tonne. The accounts need a price for the damage avoided by a tonne of CO2 that stays out of the atmosphere. Vallecillo et al. (2019) use the benchmark of EUR 30 per tonne of CO2 from OECD (2016), which they describe as a reference point and as a lower-end estimate. A decision was made to use a social cost of carbon instead, which is the value of the damage that one additional tonne

causes, because it reflects the long-term damage of climate change and does not depend on the conditions of a carbon market in a given year.

The value draws on the meta-analysis of published estimates by Tol (2023). That work standardises every estimate to 2010 US dollars per tonne of carbon, for emissions in 2010 (Tol, 2022), and its update finds a weighted mean of 207 US$ per tonne of carbon, which rises to between 231 and 253 US$ when older studies are discounted (Tol, 2025). A decision was made to adopt 266 US$ per tonne of carbon in 2010 prices, which lies just above that range and well below the kernel mean of 509 US$ in Tol (2022), whose long right tail reflects the large uncertainty of these estimates. We moved it to 2023 prices with the consumer price index of the United States (U.S. Bureau of Labor Statistics, 2023), a factor of 304.70/218.06, and to euros at the rate of 1.0813 US$ per euro in 2023 (European Central Bank, 2023). Those steps lead to about EUR 344, and the accounts use EUR 343.6 per tonne of carbon. One tonne of CO2 holds 12/44 of a tonne of carbon, so the price is EUR 93.7 per tonne of CO2 in 2023 euros (see 07_prepare_carbon_price.R in Annex I).

The social cost of carbon is a single real value, so we held it constant in 2023 euros and moved it to the prices of each year t with the harmonised index of consumer prices H of the European Union (Eurostat, 2026c) (see 03_download_hicp.R in Annex I). Constant prices of 2010 remove general inflation, so values of different years can be compared.

$$P_t = P_{2023}\,\frac{H_t}{H_{2023}}, \qquad V_{c,t}^{\text{current}} = F_{c,t}\,P_t, \qquad V_{c,t}^{\text{constant}} = V_{c,t}^{\text{current}}\,\frac{H_{2010}}{H_t} \tag{5}$$

The price moves from EUR 54.2 per tonne of CO2 in 2000 to EUR 96.1 in 2024 in current prices. The missed flow is valued in the same way (see 09_build_monetary_accounts.R in Annex I). The handbook of Eurostat does not value this service (Eurostat, 2026a).

2.7 Maps

2.7.1 The spatial key

Vallecillo et al. (2019) map forest land only, since it is the one ecosystem type for which almost all countries report uptake, and they spread the national values with dry matter productivity over the woodland and forest of Corine Land Cover. We followed that approach, with the two more recent products that the Earth observation roadmap of the same programme names (Zurbarán Nucci et al., 2025).

Forest comes from the Corine accounting layers, which the European Environment Agency prepares for comparison between years, in version 21 for all five years (European Environment Agency, 2026). The classes are broad-leaved forest (311), coniferous forest (312), mixed forest (313) and transitional woodland and shrub (324). We reduced them from 100 m to the share of each 1 km cell (see 04_prepare_clc.R in Annex I).

Productivity is the annual sum of the 36 ten-day GDMP values of the Copernicus Land Monitoring Service, each multiplied by its number of days. The 1 km product, version 2, covers 2000, 2006 and 2012 (Copernicus Land Monitoring Service, n.d.-a), and it ends in 2020. The 300 m product, version 2, covers 2018 and 2024 (Copernicus Land Monitoring Service, n.d.-b), and it starts in 2014 (see 05_prepare_gdmp.R in Annex I). Since 2018 exists in both, we compared them over 2.0 million cells that are mostly forest. The correlation is 0.967, and the mean of the 300 m product is 94.0 percent of the mean of the 1 km product. Weights only share a national value among the cells of a country, so a difference in level between the products changes no national total. The comparison was run once, and the pipeline keeps its result only when the 1 km product of 2018 is available.

The weight of a cell i is its forest share S_i times its annual productivity G_i, and each cell receives its part of the national value.

$$F_i = F_c \frac{S_i \, G_i}{\sum_{i \in c} S_i \, G_i} \tag{6}$$

2.7.2 Products

We wrote the maps at 1 km, which are a working product kept outside the repository, and at 100 m, which are the published product (see 10_write_maps_1km.R in Annex I and 11_write_maps_100m.R in Annex I). At 100 m the forest share is 1 for a forest cell and 0 otherwise, and productivity is interpolated bilinearly to the grid. There are three maps for each reporting year. The first is the actual flow, in thousand tonnes of CO2 per cell. The second is its value, in million euro at constant 2010 prices per cell. The third is the flow without the offset, for which a decision was made to map the removals of forest land, following Vallecillo et al. (2019), because other land categories switch between source and sink across years and countries. This map therefore holds less than the flow without the offset in the tables. In 2024 the difference is 34.0 million tonnes, which the tables report by region. The missed flow is a balance between removals and emissions and has no place of its own on the land, so a decision was made to report it in tables only.

2.8 Checks

The scripts stop if any check fails. Table 5 records the largest gap of each check. The sectors differ from total emissions without land use only for SE in 2022, where the inventory holds 6.1 million tonnes in the total that it does not hold in the sectors, and this year is not one of the reporting years. The total of land use, `CRF4`, equals the sum of its parts in the reporting years, with a largest difference of 0.07 thousand tonnes.

Table 5: Consistency checks, with the largest gap and the tolerance that would stop the scripts. Gaps are relative unless they are marked as absolute.

Check	Largest gap	Tolerance	Type
Shares of the forest removals add up to one in every year	1.1e-16	1e-06	Relative
National flows add up to the EU net flow in every year	1.3e-16	1e-06	Relative
Missed flow adds up to the EU emissions in every year	4.3e-16	1e-06	Relative
Flow without offset equals actual flow plus missed flow	4.5e-13	1e-09	Absolute
Sector shares add up to one in every country and year	2.2e-16	1e-06	Relative
Sectors add up to total CO2 emissions without land use	2.2e-03	5e-03	Relative
Countries add up to the EU27 aggregate of Eurostat, in kt CO2	4.0e-02	1e-01	Absolute
The deflator equals one in the base year	0.0e+00	1e-12	Relative
Constant values equal current values times the deflator	0.0e+00	1e-09	Absolute
Actual value plus missed value equals the value without offset	9.1e-13	1e-09	Absolute
NUTS2 and NUTS3 regions add up to the countries, in every table, metric and year	2.1e-16	1e-06	Relative
1 km flow maps against the tables, in every year	4.7e-11	1e-05	Relative
1 km value maps against the tables, in every year	7.1e-11	1e-05	Relative

Check	Largest gap	Tolerance	Type
1 km maps of forest removals against the tables, in every year	3.9e-11	1e-05	Relative
1 km flow maps against the tables, in every country and year	1.2e-08	1e-05	Relative
100 m maps against the tables, in every country and year	5.7e-08	1e-04	Relative
100 m maps against the tables, in every year	1.6e-08	1e-04	Relative

We read the maps back from disk to compare them with the tables. At 1 km, the forest area of the grid is 157.4 million hectares in 2024, the same as in the table of regions. At 100 m, the largest difference from the tables in any year is 1.6e-06 percent (see 12_check_maps_100m.R in Annex I). The number of countries that hold a flow and lack forest cells, which would lose that flow on the map, is 0.

3 Results

3.1 Removals, emissions and actual flow

Forest land provides 91 percent of the removals in 2024, and the other five land categories are net emitters at the level of the European Union (Table 6). The balance of forest land falls from 477.0 million tonnes in 2012 to 311.6 million tonnes in 2018, and that fall drives the series (Figure 1). Emissions change much less. As a result, the actual flow falls from 371.7 to 225.2 million tonnes between the same years, and then recovers to 236.0 million tonnes in 2024. Values of recent years may change with new releases of the inventory.

Table 6: Signed balance by category in the EU27, in million tonnes of CO2, with uptake as positive. The last two categories stay outside the flow.

Category	2000	2006	2012	2018	2024
Forest land	450.5	469.5	477.0	311.6	336.4
Cropland	-67.4	-58.6	-44.0	-34.2	-44.9

Category	2000	2006	2012	2018	2024
Grassland	-20.0	-13.0	-16.3	-4.0	-3.5
Wetlands	-12.2	-15.6	-14.3	-17.6	-14.2
Settlements	-23.1	-25.9	-29.2	-28.7	-35.9
Other land	-1.5	-1.4	-1.5	-1.9	-1.9
Harvested wood products	45.5	62.5	31.3	39.8	24.8
Other	-0.6	-0.3	-0.1	-0.1	-0.0

Table 7: Flow without offset, emissions, actual flow and missed flow in the EU27, in million tonnes of CO2.

Year	Flow without offset	Emissions	Actual flow	Missed flow
2000	470.3	144.0	326.3	144.0
2006	497.5	142.4	355.1	142.4
2012	498.5	126.9	371.7	126.9
2018	366.8	141.6	225.2	141.6
2024	375.0	139.1	236.0	139.1

3.2 Countries

Under Equation 3, the actual flow of a country follows its forest removals, so the flow is concentrated. Five countries hold 77 percent of the actual flow in 2024, led by FR, IT, SE, ES, PL (Table 8 and Figure 2). In 2024 3 countries have no forest removals, and in 2018 there are 5, whereas no country is in that position in 2000, 2006 or 2012. These countries have no actual flow in those years. Germany held 36.7 million tonnes in 2012, and it holds none in 2018 or 2024.

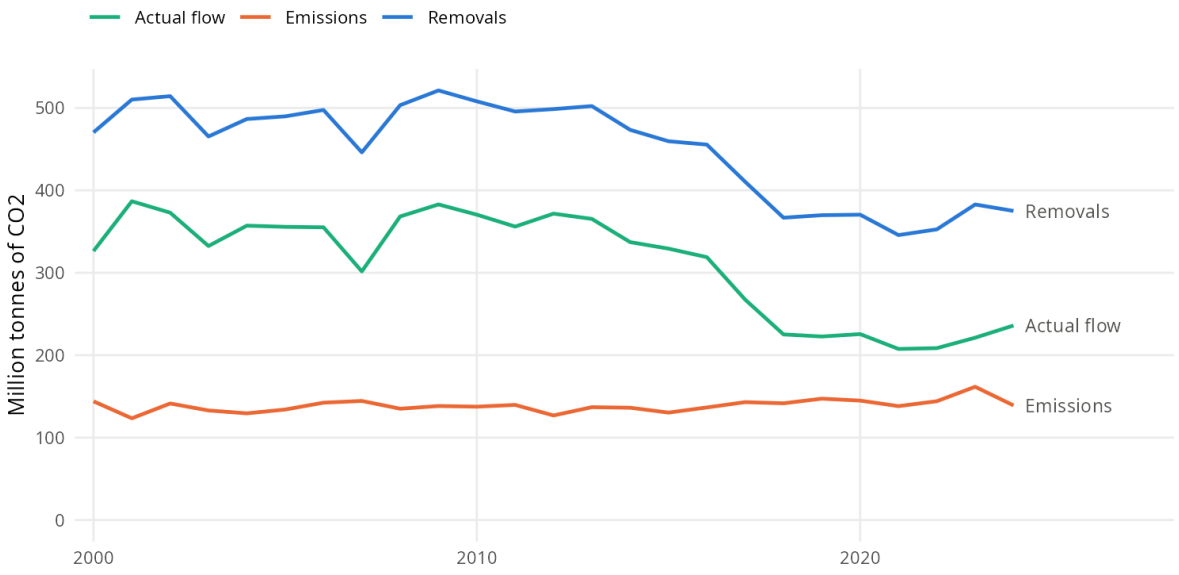


Figure 1: Removals, emissions and actual flow in the EU27, 2000 to 2024.

Table 8: Accounts by country in 2024, in thousand tonnes of CO2, with values in million euro at constant 2010 prices.

Country	National balance	Flow without offset	Actual flow	Missed flow	Value of actual flow	Value of missed flow
FR	43,208	65,257	44,514	20,742	3,071	1,431
IT	62,608	63,734	40,060	23,674	2,763	1,633
SE	52,078	56,447	38,738	17,709	2,672	1,222
ES	50,676	52,166	33,129	19,037	2,285	1,313
PL	23,844	38,739	24,531	14,208	1,692	980
RO	31,149	36,694	16,998	19,697	1,172	1,359
BG	7,644	8,610	5,635	2,976	389	205
LT	6,097	7,651	4,962	2,689	342	185
HU	5,994	6,436	4,421	2,015	305	139
HR	4,168	4,936	3,256	1,680	225	116
SI	4,149	4,462	2,975	1,487	205	103

Country	National balance	Flow without offset	Actual flow	Missed flow	Value of actual flow	Value of missed flow
SK	4,499	4,704	2,893	1,811	200	125
DK	603	3,543	2,452	1,091	169	75
CZ	3,376	3,738	2,017	1,721	139	119
IE	535	2,854	1,975	879	136	61
NL	-2,852	2,435	1,685	750	116	52
FI	-9,531	2,154	1,491	663	103	46
BE	692	2,042	1,384	658	95	45
EL	4,609	4,886	1,218	3,668	84	253
LV	-2,711	857	593	264	41	18
EE	-2,286	997	512	484	35	33
LU	632	664	402	261	28	18
CY	191	206	134	72	9	5
MT	-1	5	0	5	0	0
AT	-5,230	0	0	0	0	0
DE	-47,799	0	0	0	0	0
PT	-365	810	0	810	0	56

3.3 Regions

The workbook holds every quantity at NUTS0, NUTS2 and NUTS3. Table 9 lists the NUTS2 regions with the largest actual flow in 2024, which are the large forest regions of the countries with the highest removals.

Table 9: NUTS2 regions with the largest actual flow in 2024, in thousand tonnes of CO2, with the value in million euro at constant 2010 prices.

Code	Region	Actual flow	Flow without offset	Missed flow	Value of actual flow
SE33	Övre Norrland	10,206	14,938	4,732	704

Code	Region	Actual flow	Flow without offset	Missed flow	Value of actual flow
SE31	Norra Mellansverige	7,796	11,276	3,480	538
SE32	Mellersta Norrland	6,431	9,336	2,905	444
FRI1	Aquitaine	5,710	8,290	2,580	394
ES41	Castilla y León	5,513	8,838	3,325	380
ITI1	Toscana	5,232	7,900	2,667	361
FRK2	Rhône-Alpes	4,792	7,010	2,218	331
SE12	Östra Mellansverige	4,624	6,723	2,099	319
SE21	Småland med öarna	4,568	6,675	2,106	315
ES11	Galicia	4,436	6,556	2,120	306

3.4 Maps

Figure 3 shows the actual flow, drawn at 10 km so that it reads at this size. Forest regions of high productivity stand out within each country, and the differences between countries reflect their national values, so borders can show as steps in the map. Vallecillo et al. (2019) note the same effect. Countries with no forest removals in a year appear blank in that panel, as Germany does in 2018 and 2024. The mean productivity on forest rises from 23,500 to 26,664 kilograms per hectare between 2012 and 2018, although the 300 m product runs below the 1 km product in the year they share, so the change of product does not explain the rise. Productivity only places values within a country, so this has no effect on national totals. The value maps in Figure 4 follow the same pattern, since they multiply the flow by one price per year.

3.5 Value

The actual flow is worth 16.3 billion euro in 2024 at constant 2010 prices, and 22.7 billion euro in current prices (Table 10 and Figure 5). Constant values follow the physical flow, because the real price is fixed. The missed flow is worth between 8.8 and 9.9 billion euro in the reporting years, since emissions change little.

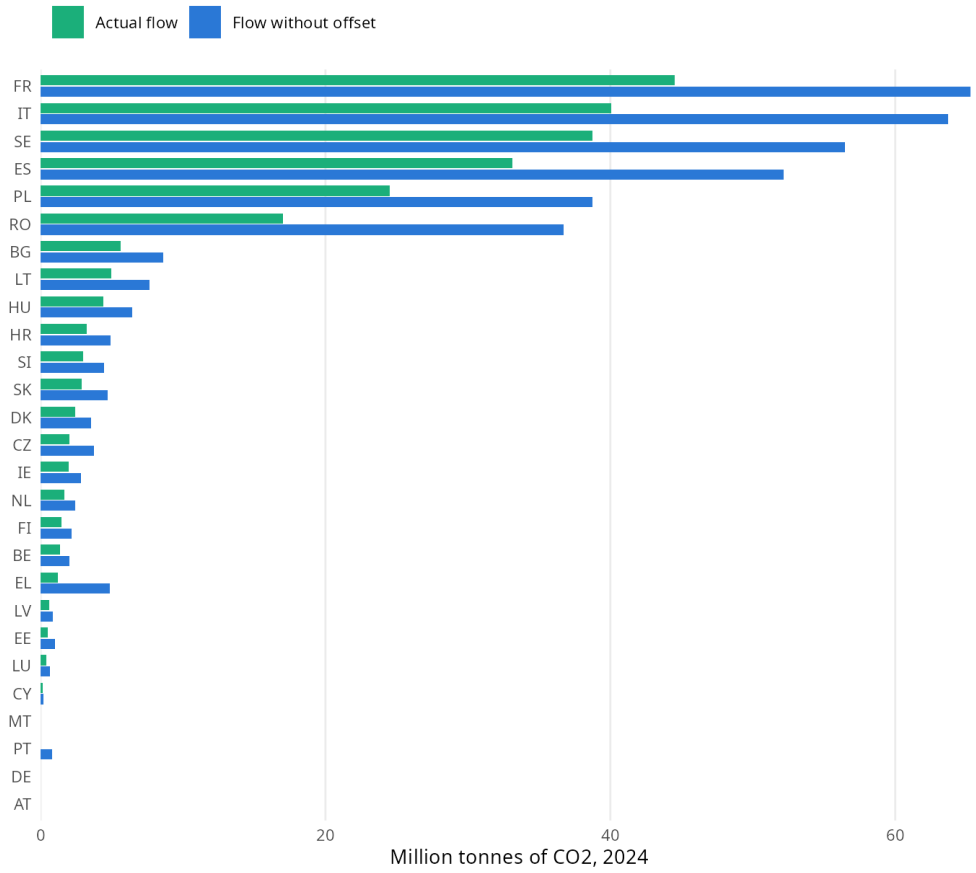


Figure 2: Actual flow and flow without offset by country, 2024.

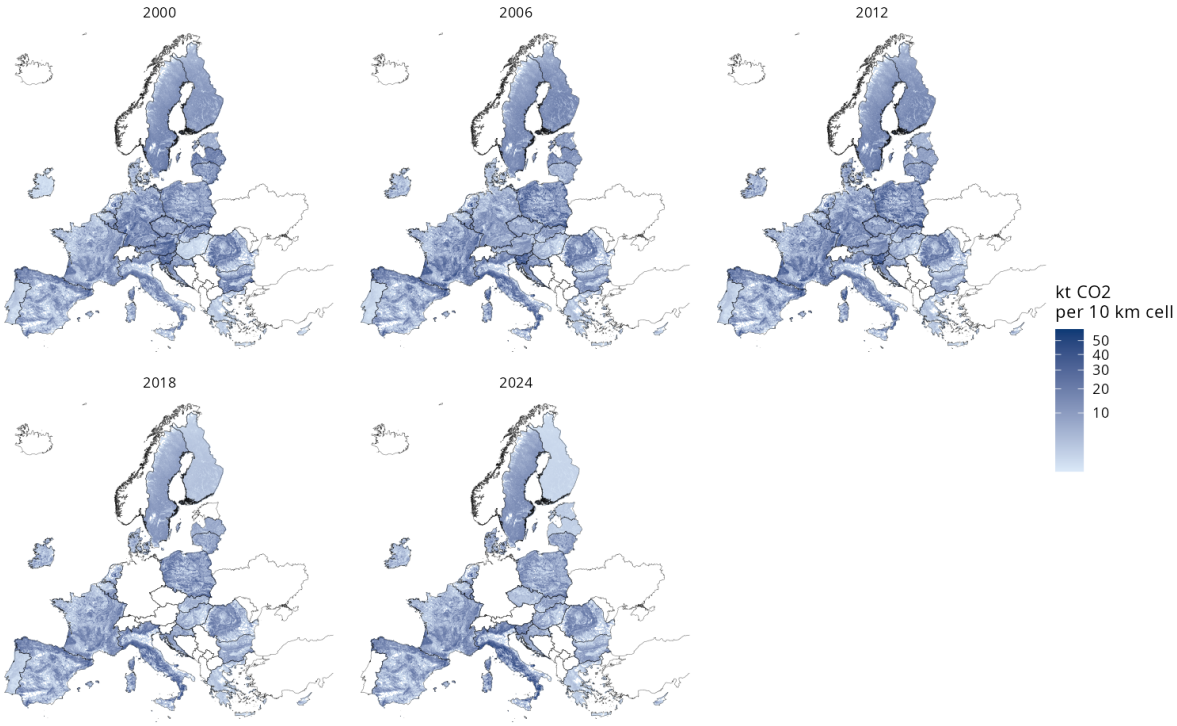


Figure 3: Actual flow at 10 km, in thousand tonnes of CO2 per cell, on a square root scale.

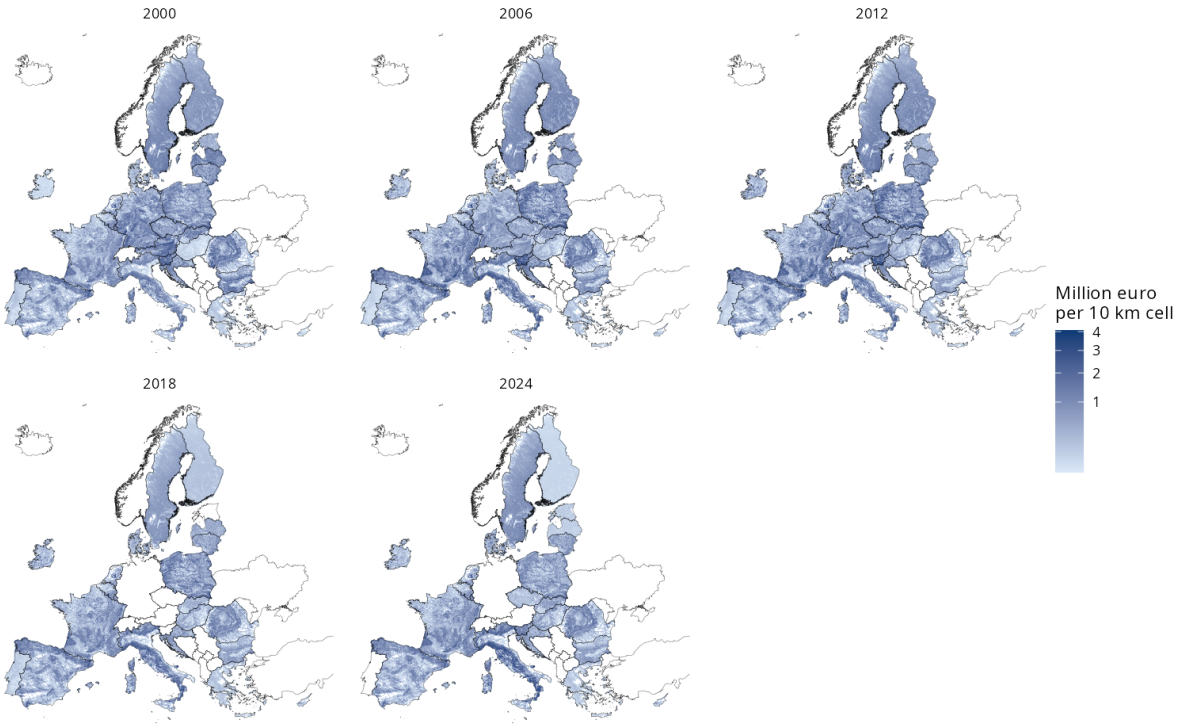


Figure 4: Value of the actual flow at 10 km, in million euro at constant 2010 prices per cell, on a square root scale.

Table 10: Value in the EU27, in million euro.

	Year	Actual flow, current prices	Actual flow, constant 2010 prices	Missed flow, constant 2010 prices
	2000	17,695	22,510	9,933
	2006	22,483	24,498	9,820
	2012	27,061	25,639	8,751
	2018	17,292	15,532	9,768
	2024	22,689	16,278	9,592

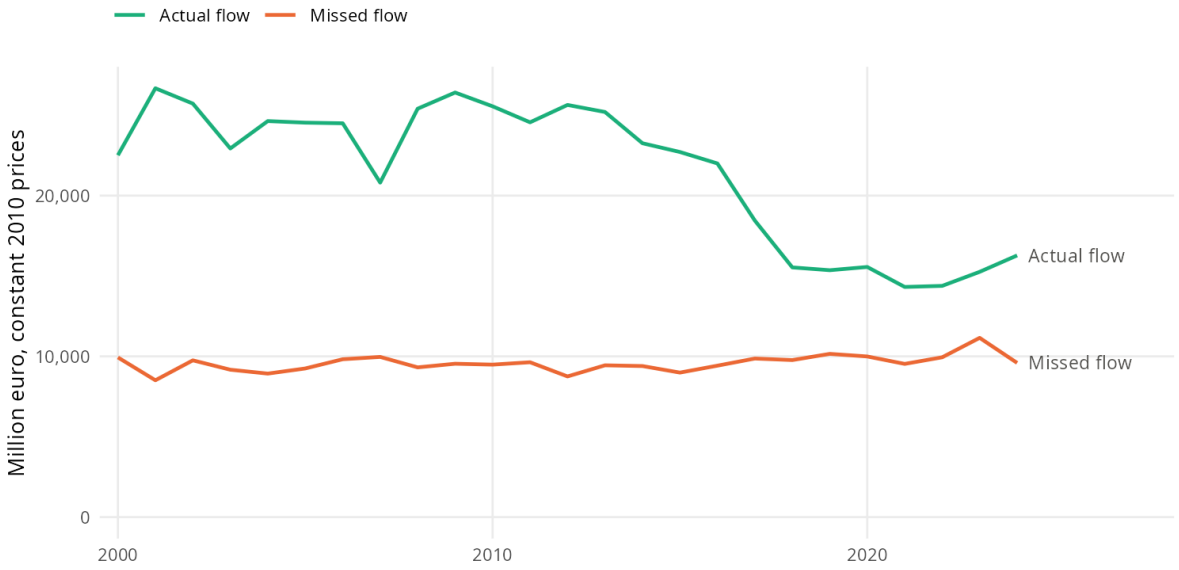


Figure 5: Value of the actual flow and of the missed flow in the EU27, at constant 2010 prices.

3.6 Sectors

Table 11 shows the actual flow attributed to the sectors that emit CO2. The largest part goes to energy supply in 2012 and to transport in 2024. The attribution follows the emissions of each country weighted by its flow, so it also shifts when the flow moves between countries.

Table 11: Actual flow attributed to emitting sectors in the EU27, in million tonnes of CO2.

Sector	2000	2006	2012	2018	2024
Transport	77.4	89.8	101.0	71.5	92.0
Energy supply	111.9	125.0	128.7	65.4	52.8
Manufacturing and construction	85.7	85.3	83.1	51.7	52.8
Households	29.4	30.6	32.8	19.8	19.2
Services and other	12.9	15.2	15.3	9.4	10.3
Agriculture, forestry and fishing	8.7	9.0	10.4	7.1	8.6
Waste	0.2	0.4	0.4	0.3	0.3

3.7 Comparison with the published accounts

La Notte et al. (2021) report an actual flow of 306 million tonnes of CO2 in 2012 in their main tables, and 303 million tonnes in one passage of the text, from forest removals of 444 million tonnes. We obtain 371.7 million tonnes (Table 12). The comparison serves for orientation only, because they report the EU-28 with an earlier release of the inventory, and countries revise past years in each release. The price also differs by design. Their flow of 2012 is valued at 9,189 million euro, which implies EUR 30.0 per tonne, whereas the price here in 2012 is EUR 72.8 per tonne in current prices.

Table 12: Actual flow in this report (EU27) and in La Notte et al. (2021), Table 6.7 (EU-28), in million tonnes of CO2.

Year	This report	Published	Difference (%)
2000	326.3	291.6	11.9
2006	355.1	292.2	21.5
2012	371.7	306.3	21.3

4 Limitations

Inventories. Countries produce their inventories with their own methods and definitions, so comparisons between countries call for caution (Vallecillo et al., 2019). They revise them every year, which changes the values of past years between releases. Some sharp changes between years may reflect these methods as well as forests, and we did not separate the two.

Actual flow. The flow of a country depends on the removals and emissions of all countries, because the offset is computed once for the European Union, and it should not be read as a national balance. The tables show that balance separately. A category counts as a removal or an emission according to its net sign in each country and year, so the gross fluxes inside a category are not visible. Countries with no forest removals in a year receive no flow, so their value can change abruptly between years when their forest moves between source and sink. Harvested wood products stay outside the flow, and including them as a deduction would lower the removals of forest land by up to the sink in Table 4.

Maps. The spatial key combines where forest is located with the productivity of the vegetation, which is a proxy for where removals happen. Productivity measures growth and does not measure the net carbon balance, so it does not capture the carbon in soils or the age of stands. The maps of 2000, 2006 and 2012 use a 1 km product and those of 2018 and 2024 a 300 m product, so the spatial detail of the two groups differs. The Corine accounting layers have a minimum mapping unit of 25 hectares. Maps and regional values allocate a national value, so they should not be read as independent local estimates. The map of the flow without the offset covers forest land only.

Value. The value depends directly on the price. The social cost of carbon is a model-based estimate that is sensitive to assumptions such as the discount rate and the damage function. This literature quotes estimates per tonne of carbon and per tonne of CO2, which differ by a factor of 44/12, and the value adopted here is per tonne of carbon, the unit in which the meta-analysis standardises its estimates. The real price is about 3 times the benchmark of earlier accounts, so the values cannot be compared with those accounts. It also differs from a market price. A consumer price index for the whole European Union converts values to constant prices, so differences in inflation between countries are ignored.

References

Copernicus Land Monitoring Service. (n.d.-a). *Gross dry matter productivity 1999-2020 (raster 1 km), global, 10-daily, version 2.* Copernicus Land Monitoring Service. https://doi.org/10.2909/b54be6e3-5962-4e9a-a443-9ff7ea943dfe

Copernicus Land Monitoring Service. (n.d.-b). *Gross dry matter productivity 2014-present (raster 300 m), global, 10-daily, version 2.* Copernicus Land Monitoring Service. https://doi.org/10.2909/fd838f95-21e9-4f37-a9b8-b438d9dc4edf

European Central Bank. (2023). *Euro foreign exchange reference rates, US dollar, annual average 2023*. Retrieved from https://data.ecb.europa.eu/currency-converter

European Environment Agency. (2026). *Corine land cover accounting layers (raster 100 m), version 21*. EEA Datahub. https://doi.org/10.2909/a65428d3-bd66-4b47-bcf9-6bfd6fa5df0c

Eurostat. (2026a). *Ecosystem accounts handbook, 2026 edition*. Luxembourg: Publications Office of the European Union.

Eurostat. (2026b). *Greenhouse gas emissions by source sector (source: EEA) [env_air_gge]*. Eurostat database. https://doi.org/10.2908/ENV_AIR_GGE

Eurostat. (2026c). *HICP, annual data (average index and rate of change) [prc_hicp_aind]*. Eurostat database. https://doi.org/10.2908/PRC_HICP_AIND

Eurostat. (2026d). *Volume of timber over bark (EFA questionnaire) [for_vol_efa]*. Eurostat database. https://doi.org/10.2908/FOR_VOL_EFA

Eurostat GISCO. (2024). *Administrative units, statistical units: NUTS 2024, scale 1:1 million*. Retrieved from https://ec.europa.eu/eurostat/web/gisco/geodata/statistical-units/territorial-units-statistics

La Notte, A., Vallecillo, S., Garcia-Bendito, E., Grammatikopoulou, I., Czúcz, B., Ferrini, S., … Maes, J. (2021). *Ecosystem services accounting. Part III: Pilot accounts for habitat and species maintenance, on-site soil retention and water purification* (No. EUR 30856 EN, JRC126566). Publications Office of the European Union.

OECD. (2016). *Effective carbon rates: Pricing CO2 through taxes and emissions trading systems*. Paris: OECD Publishing.

Tol, R. S. J. (2022). *Estimates of the social cost of carbon have increased over time*. Retrieved from https://arxiv.org/abs/2105.03656

Tol, R. S. J. (2023). Social cost of carbon estimates have increased over time. *Nature Climate Change*, *13*, 532–536. https://doi.org/10.1038/s41558-023-01680-x

Tol, R. S. J. (2025). Trends and biases in the social cost of carbon. *Annals of the New York Academy of Sciences*, *1548*(1), 248–259. https://doi.org/10.1111/nyas.15340

U.S. Bureau of Labor Statistics. (2023). *Consumer price index for all urban consumers (CPI-U), U.S. City average, all items*. Retrieved from https://www.bls.gov/cpi

Vallecillo, S., La Notte, A., Kakoulaki, G., Kamberaj, J., Robert, N., Dottori, F., … Maes, J. (2019). *Ecosystem services accounting. Part II: Pilot accounts for crop and timber provision, global climate regulation and flood control* (No. EUR 29731 EN, JRC116334). Publications Office of the European Union.

Zurbarán Nucci, M., Toro Herrera, J. F., Vallecillo, S., La Notte, A., Zulian, G., Polce, C., … Paracchini, M.-L. (2025). *Earth observation roadmap for ecosystem services accounting in the EU* (No. JRC142634). Joint Research Centre, European Commission.

Annex I. Scripts of the pipeline

This annex describes each script in the order in which it runs. The link under each name opens its code on GitHub, and the paragraph explains what the script takes in, what it does and what it saves for the steps that follow. Scripts 01 to 07 prepare the inputs, and several of them need large source files that are kept outside the repository. Scripts 08 to 10 run from the repository alone. Scripts 11 and 12 write and check the 100 m maps, and they need the large source files again.

01_prepare_nuts.R

View on GitHub

This script takes the NUTS 2024 boundaries of Eurostat at a scale of 1:1 million (Eurostat GISCO, 2024), in the copy distributed with the INCA plugin, and keeps only the level of countries. It marks the 27 Member States and stops if it does not find all of them. The result is a small file, `NUTS_RG_01M_2024_3035_LEVL0.gpkg`, which the later scripts use to identify the countries and to draw the maps. The repository holds the source file with its copyright notice.

02_download_eurostat.R

View on GitHub

This script downloads the CO2 emissions and removals by source sector from the Eurostat dataset `env_air_gge` (Eurostat, 2026b) through the Eurostat data service. It keeps the values in thousand tonnes for the 27 Member States and the EU27 aggregate from 1990 to 2024, drops the rows with no reported value and records the date of the download. The table is saved as `env_air_gge_co2.rds` for the physical accounts. A new release of the dataset changes the results, so this step runs only when an update is intended.

03_download_hicp.R

View on GitHub

This script downloads the annual harmonised index of consumer prices of the EU27, for all items and with 2015 as its base (Eurostat, 2026c). The series is saved as `hicp_eu27.rds`, which the carbon price and the constant prices of the monetary accounts use.

04_prepare_clc.R

View on GitHub

This script reads the Corine Land Cover accounting layers at 100 m for the five reporting years (European Environment Agency, 2026) and groups their classes into the six land categories of the inventory, leaving water bodies out. It reduces each layer to the share of every 1 km cell covered by each category. The forest share of each year is saved as `CLC<year>_forest_pct_1km.tif`, in whole percentages, which brings a layer of about 175 MB down to a file of under 4 MB. The script then adds up the hectares of each category in every country and in every NUTS2 and NUTS3 region, and it saves them as `clc_ecosystem_area_by_nuts.rds`. The later scripts use these files to weigh regions and cells.

05_prepare_gdmp.R

View on GitHub

This script builds the annual productivity that places the flow on the map. For each year it reads the 36 ten-day files of Gross Dry Matter Productivity from the Copernicus data store, using the 1 km product for 2000, 2006 and 2012 (Copernicus Land Monitoring Service, n.d.-a) and the 300 m product for 2018 and 2024 (Copernicus Land Monitoring Service, n.d.-b). It keeps the part of each file that covers Europe, multiplies each daily rate by the number of days it represents, counts missing cells as zero and adds up the year. This annual file stays outside the repository at its original resolution for the 100 m maps, and a copy on the 1 km grid of the accounts is saved as `GDMP_<year>_1km.tif`. Because 2018 exists in both products, the script also compares them over the cells that are mostly forest and saves the result as `gdmp_comparison.rds` for the report. The store requires a free Copernicus account only when an annual file has to be built again.

06_prepare_nuts_weights.R

View on GitHub

This script works out the share of each NUTS2 and NUTS3 region in the national total of each land category. The weight of forest is the forest share of each 1 km cell times its productivity in the same year, added up over the region, and the weight of the other categories is their area in hectares. When a country has no mapped area of a category, the share of the region in the total mapped land of the country stands in, and the row is marked. The shares are saved as `clc_type_weight_by_nuts.rds`, which the monetary accounts use to build the regional results.

07_prepare_carbon_price.R

View on GitHub

This script sets the social cost of carbon at EUR 343.6 per tonne of carbon in 2023 euros, based on Tol (2023), and converts it to tonnes of CO2 with the ratio of 12/44. It holds that value constant in real terms and moves it to the prices of each year from 2000 to 2024 with the consumer price index. The series is saved as `carbon_price.rds` for the monetary accounts.

08_build_physical_accounts.R

View on GitHub

This script builds the physical accounts for every year from 2000 to 2024, following Vallecillo et al. (2019) and La Notte et al. (2021). It reverses the sign of the inventory, separates the removals and emissions of each land category in each country and computes the net flow once for the European Union. It attributes that flow to countries in proportion to their removals of forest land, and it derives the flow without the offset and the missed flow. It then shares the flow of each country among the sectors that emit CO2. The script checks that countries add up to the Eurostat aggregate, that shares add up to one and that the flows agree with each other, and it stops if any check fails. The results are saved as `carbon_lulucf_country_year.rds`, `carbon_flow_country_year.rds` and `carbon_sector_allocation_country_year.rds`, together with the small tables and the figures of the report.

09_build_monetary_accounts.R

View on GitHub

This script multiplies the physical flows by the carbon price of each year to obtain their value in current prices, and it applies the consumer price index to express them in constant 2010 prices. For the reporting years it then shares the national values among NUTS2 and NUTS3 regions, and it stops if the regions do not add up to their countries. The national values are saved as `carbon_value_country_year.rds`, and the tables by NUTS level form the workbook `carbon_sequestration_results.xlsx` on the Downloads page. The small tables and the value figure of the report are saved as well.

10_write_maps_1km.R

View on GitHub

This script places the national values on a 1 km grid for each reporting year. The weight of a cell is its forest share times its productivity, and each cell receives the national value in

proportion to its weight within its country. It writes maps of the actual flow, of its value at constant 2010 prices and of the removals of forest land, which stand for the flow without the offset. These maps are a working product kept outside the repository. The script reads them back from disk, stops if they disagree with the tables, and saves the small tables of the checks and the two map figures of the report, drawn at 10 km.

11_write_maps_100m.R

View on GitHub

This script writes the published maps at 100 m, using the same method and the national values of the tables. It works directly from the Corine accounting layers, where a cell counts as forest or not, and from the annual productivity of script 05, interpolated to the 100 m grid. It writes three maps for each reporting year, which make 15 GeoTIFF files named `carbon_flow_gdmp_100m_<year>.tif`, `carbon_value_constant_gdmp_100m_<year>.tif` and `carbon_flow_no_offset_gdmp_100m_<year>.tif`. These files are too large for the repository, so they are published in the external folder linked from the Downloads page.

12_check_maps_100m.R

View on GitHub

This script reads the 15 maps at 100 m back from disk, adds up their cells by country and compares the totals with the national tables. It stops if a country or a year differs by more than 0.01 percent. Because the maps stay outside the repository, the script saves this record as `maps_100m_totals.rds` and `checks_maps_100m.rds`, and the report reads it from there.

run_all.R

View on GitHub

This script runs the numbered scripts in order, grouped into four stages, which are the inputs (scripts 01 to 07), the tables (08 and 09), the 1 km maps (10) and the 100 m maps (11 and 12). With no stage named it runs the tables and the 1 km maps, which need only the repository.

Helper files

View on GitHub: helpers_common.R, helpers_carbon.R, helpers_monetary.R, helpers_results.R and helpers_report.R

These files hold the steps that several scripts share, and they do not run on their own. The first holds the common work with grids, regions, saved tables and checks. The second holds the

formulas of the physical accounts and the style of the figures. The third prepares the deflator of the constant prices, the fourth builds the regional tables and the workbook, and the fifth formats the tables of this report.